



Mother-of-pearl

△ **Black pearl** in its shell of mother-of-pearl

Mother-of-pearl is the name given to nacre, the substance that lines the interior of some molluscs, but chiefly pearl oysters and freshwater pearl mussel shells. It is also the material from which pearls are formed. Prized for its iridescence, nacre is used decoratively in jewellery, clothing, architecture, and art. It is also of great interest to scientists due to its overlapping brick-like structure at a microscopic level, which allows impacts to be absorbed by dispersing them over a wide area – a property that could be applied to the creation of resilient materials.

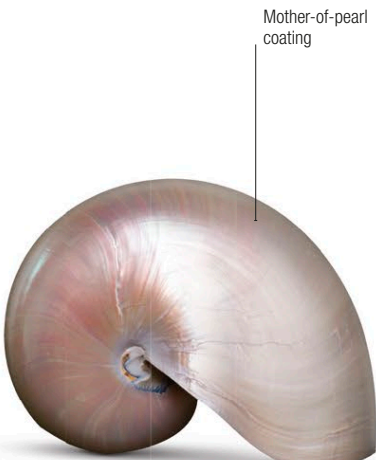
Specification

Chemical name Calcium carbonate, calcium phosphate, amorphous silica | **Formula** CaCO_3 , $\text{Ca}_3(\text{PO}_4)_2$, SiO_2 | **Colours** All | **Structure** Amorphous (prismatic, nacreous, crossed-lamellar, foliated and homogeneous) | **Hardness** 3.5 | **SG** 2.70–2.89 | **RI** 1.530–1.685 | **Lustre** Greasy to pearly | **Locations** Worldwide



Patterned shell

Turban shell | **Rough** | South African turban shells such as this have been used in tribal art for millennia, and also occasionally as currency.



Mother-of-pearl coating

Nautilus shell | **Rough** | This stunning example of a large nautilus shell is coated with a fine layer of iridescent mother-of-pearl.



Natural iridescence

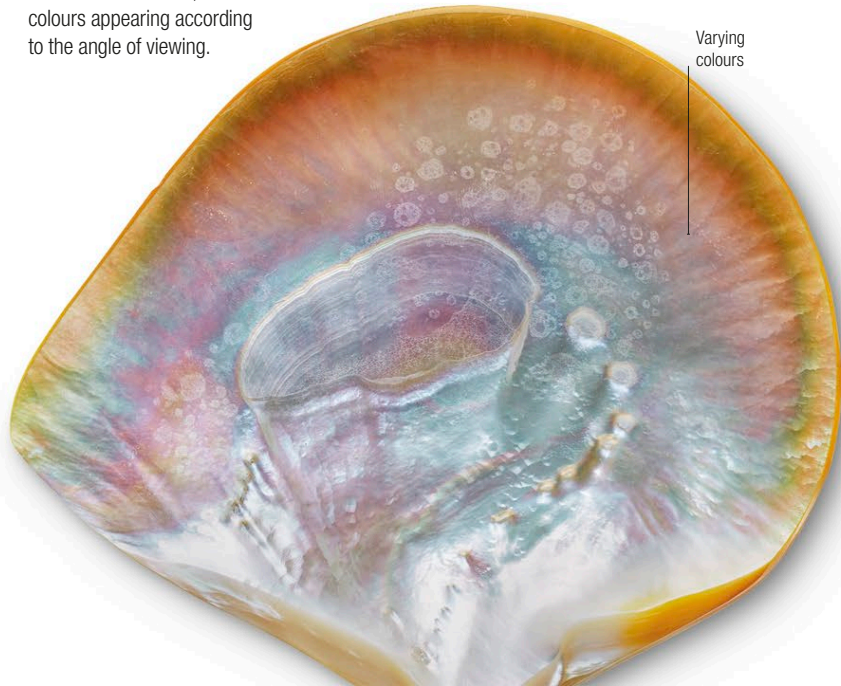
Mother-of-pearl beads | **Cut** | These flat oval beads show the natural shimmer and delicate finish that make mother-of-pearl such a popular decorative material.



Mother-of-pearl inlay

Mother-of-pearl pendant | **Set** | This fanciful pendant in the shape of a window consists of mother-of-pearl, synthetic sapphire, and diamond.

Nacreous shell | **Rough** | This shell displays excellent iridescence, with different colours appearing according to the angle of viewing.



Varying colours

Art in Asia

Mother-of-pearl and lacquer

In various Asian cultures between the 8th and 19th centuries, craftsmen produced exquisite lacquered mother-of-pearl decorative pieces. These ranged from small boxes to large screens, and featured designs illustrating various religious or cultural themes. Nacreous shells were boiled and cut into pieces to form the designs. The craftsmen then coated the pieces with many layers of lacquer, a form of resinous tree sap that hardens to a protective, plastic-like finish.



Lacquer panel This Shibayama (inlay) piece features mother-of-pearl figures of a man and boy in rural Japan.